

RUTUJA DESHMUKH

Aspiring AI Engineer | Building AI Applications for Real-World Problems

Diploma in AI & ML '26 @ KKW Polytechnic | B.E. AI & Data Science '29 @ MET

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[Portfolio](#) • [GitHub](#) • [LinkedIn](#)

ABOUT ME

I am in the second year of a B.E. in Artificial Intelligence & Data Science, after completing a **3-year Diploma in Artificial Intelligence & Machine Learning** — so my AI journey began well before engineering, and B.E. is the next step in it. I learn best by building, debugging and deploying: projects, hackathons, research and technical competitions, an internship and a client-based project. My focus is **Generative AI, RAG, multi-agent systems** and intelligent workflows that turn ideas into working AI applications.

EDUCATION JOURNEY

FOUNDATION Diploma in AI & ML K. K. Wagh Polytechnic, Nashik 3-year programme • completed 2026 • 89.76%	NOW • 2ND YEAR B.E. AI & Data Science Mumbai Educational Trust (MET), MET League of Colleges	EXPECTED Graduation: 2029 Building on the diploma foundation with hands-on projects
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SSC, Shree Ram Vidhyalaya, Nashik — 92.80% (May 2023).

CORE TECHNICAL SKILLS

Hands-on experience and working knowledge, gained by building the projects below.

AI / GENERATIVE AI

Generative AI RAG LLM Integration LangChain LangGraph Prompt Engineering Semantic Search
Vector Embeddings Conversational AI AI Agents Multi-Agent Systems

PROGRAMMING / BACKEND

Python SQL JavaScript
FastAPI Django
REST APIs

DATABASES / AI STORAGE

PostgreSQL
Supabase / pgvector SQLite
ChromaDB

FRONTEND / APPLICATIONS

Next.js React.js
Streamlit Tailwind CSS

TOOLS / PLATFORMS

Git GitHub VS Code
Hugging Face Groq
Gemini API

FEATURED PROJECTS

01 AI Multi-Agent Research & Engineering Workspace

My main recent AI project: a deployed, SaaS-style workspace where specialised agents collaborate on research, engineering, planning and critique.

- **Multi-agent design:** Research, Engineering, Planner and Critic agents, supervised by a QA Controller, orchestrated with LangGraph and LangChain.
- **Intelligent routing:** task and relevance routing decides which agent(s) should handle a request, so answers stay on topic.
- **Memory & storage:** session and memory handling through a FastAPI service with PostgreSQL / Supabase pgvector.
- **Interface:** Next.js frontend with real-time SSE streaming, Zustand state, Framer Motion and markdown rendering via remark-gfm.

Next.js FastAPI LangGraph LangChain Groq / LLMs PostgreSQL Supabase pgvector Zustand
SSE streaming Framer Motion remark-gfm

[GitHub Repository](#) (public)

02 AdaptEd AI — Adaptive Learning Platform

Built by Team SkillMatix at the Tech Fusion 24H Hackathon (Smart Education): learning that adapts to the student instead of following a fixed plan.

- **Learner model:** tracks topic mastery with an exponential-moving-average update and recommends one explainable *Next Best Learning Action* after each interaction.
- **Adaptive RAG tutor:** chat over the student's own uploaded PDFs (PyMuPDF, ChromaDB, Sentence Transformers) with source citations and six tutor personas; quizzes adjust to mastery.
- **Learning path and extras:** prerequisite-based topic locking, coding challenges, a Mermaid diagram studio and a parent observer portal.
- **My role:** backend development and AI integration within the team.

Next.js TypeScript FastAPI PostgreSQL / Supabase SQLite ChromaDB PyMuPDF Sentence Transformers
Google Gemini API JWT

[GitHub Repository](#) (public)

03 AI Learning Companion

A deployed RAG-based study assistant that turns uploaded documents into a personalised learning workspace.

- **Document understanding:** upload PDFs/documents (PyMuPDF) and chat with them through a hybrid RAG pipeline using ChromaDB and Sentence Transformers.
- **Personalised learning:** summaries, quiz generation, coding and interview questions, project recommendations, roadmap tracking, weak-topic and learning analytics.
- **Five learning modes:** Beginner, Exam, Practical, Interview and Cheat Sheet; plus a voice assistant and Mermaid.js diagrams.

Python Streamlit ChromaDB PyMuPDF Sentence Transformers Groq Vector Embeddings

[Live Demo](#) • [GitHub Repository](#) (public)

04 AgroBot — AI-Powered Smart Robotic Lawn Mower

My final-year diploma project: a software + hardware system for autonomous grass mowing.

- **My contribution:** software development and Raspberry Pi integration, including obstacle-response logic and remote commands from an Android app over a local network.
- **Features:** GPS/maps, mission planning, remote control, obstacle alerts, and mission/history tracking.
- **Recognition:** presented at technical competitions and exhibitions, including **IET Expo 2026**; I contributed to the related research paper published in **IJARCCCE**.

Python Raspberry Pi Node.js IoT Android Integration

[Research Paper — IJARCCCE](#) • [GitHub: Mower Software](#) • [GitHub: Android App](#) (public)

MORE PROJECTS

SignalIQ — a market and competitive-intelligence dashboard that scores signals by recency, velocity and relevance, with action-first insight cards, a competitor radar and a feedback-weighted recommendation engine. React, Vite, Recharts; runs on curated seed data with localStorage, live data sources planned. Built for the Raj Shamani × Emergent Builder's Challenge. [GitHub](#)

Scriptora — a collaborative script-writing and project-management workspace; a commissioned project in active development (Next.js, FastAPI, PostgreSQL, Tiptap editor). Features are being rolled out in phases. [GitHub](#)

EcoCart — a Django e-commerce web application with product catalogue, cart and order flow, built during my internship. [GitHub](#)

AI-Based Energy Monitoring System — a Django platform with IoT sensor data, a data-processing pipeline and an analytics dashboard, built during my internship.

INTERNSHIP EXPERIENCE

Python Application Development & Data Engineering Intern

ProWorld Technology Pvt. Ltd., Nashik • 3 months, June–August 2025

- Built Django-based web applications, with UI work in HTML, CSS and Django templates.
- Worked on data handling, debugging, testing and usability within a team development workflow.
- Contributed to two projects, the **AI-Based Energy Monitoring System** and the **EcoCart** e-commerce website (see More Projects).

Django

Python

HTML/CSS

REST APIs

JavaScript

IoT data

CURRENT CLIENT PROJECT — ENGINEERX

EngineerX — for AnaVruksha (private repository)

An AI-powered engineering workspace that solves real-world hardware engineering problems through intelligent agent workflows.

- **V1 focus:** engineering assistance for rover systems such as motor control and the power subsystem, with room to extend across mechanical, electronics, electrical and robotics work.
- **Planned workflows:** circuit recommendations, simulation assistance, BOM generation, cost estimation, supplier recommendations, embedded-code and ROS-related assistance, mechanical recommendations and validation.
- **Tools being explored:** AI agents, LLMs, and engineering tools such as Fusion 360, KiCad, Gazebo and Isaac Sim.
- **My involvement:** understanding client requirements, communicating technical ideas, coordinating tasks and managing project responsibilities as the work requires.

Ongoing project, not yet deployed. The repository is private, so no public link is available.

HACKATHON & RESEARCH

Tech Fusion 24H Hackathon 2026

IT Dept., MET Bhujbal Knowledge City • 17–18 Sept 2026

- **Problem:** Smart Education — Personalized AI Learning Assistant.
- **Team SkillMatix** (4 members); my role: backend and AI integration.
- Built **AdaptEd AI** in 24 hours; participated, no placement claimed. It strengthened fast execution, collaboration and problem solving under pressure.

Avishkar Research Competition 2026–27

Engineering & Technology • research idea presentation

“AI-Based Early Prediction of Food Spoilage Risk Using Environmental and Storage Parameters Without Specialized Hardware.”

- Concept: an AI/ML decision-support approach using temperature, humidity, food type, storage duration and storage condition.
- Presented as a research idea; no prototype or results are claimed.

LEADERSHIP, TEAMWORK & COMMUNICATION

- **Teamwork and ownership:** contributed in team hackathons and project builds, taking responsibility for backend and AI integration.
- **Coordination:** client communication and task coordination on the EngineerX project, while managing deadlines.
- **Presenting:** presented AgroBot at technical competitions and exhibitions and an original research idea at Avishkar; used to explaining technical concepts to others.
- **Debate: First Prize** in the 2026 debate competition at MET Institute of Engineering (topics included Technology & Agriculture and Democracy vs Dictatorship).

CERTIFICATIONS

- Introduction to Generative AI — Google Cloud & Simplilearn
- Build AI Agents using LangGraph — Simplilearn
- AWS Academy Graduate, Cloud Foundations — Amazon Web Services
- Big Data Analytics Projects with Apache Spark — Infosys Springboard
- Database Management System — Infosys Springboard
- Python Training — Spoken Tutorial, IIT Bombay

KEY STRENGTHS

Problem Solving Debugging and breaking down tasks, from 24-hour hackathon builds to multi-agent workflows.	Technical Curiosity Learns by building, from RAG pipelines to Raspberry Pi hardware.	Ownership Takes responsibility for backend, integration and deployment of what I build.
Team Collaboration Team hackathons, internship workflow and a client project.	Communication Presents projects and research; first prize in debate.	Fast Learning Moved from diploma-level ML to GenAI, agents and deployment.

LINKS & CONTACT

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